
Reconstructing Microscopy Images from Drift-Degraded Z-stacks

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Abstract

Microscopy imaging often relies on z-stack acquisition followed by projection into a single 2D image. However, physical factors such as stage drift and focus drift degrade image quality, leading to spatial misalignment and defocus blur. In this work, we simulate physically motivated degradation processes and train a U-Net model to reconstruct clean fused images from distorted z-stacks. Specifically, we model XY drift as cumulative random translations and Z drift as depth-dependent Gaussian blur. A small U-Net is trained on patch-based data using a combination of L1 and edge-aware losses. Results show significant improvement in structural similarity (SSIM from ~ 0.51 to ~ 0.95), while high-frequency details are only partially recovered. These findings highlight both the effectiveness and limitations of learning-based reconstruction under simplified physical simulations.

1 Introduction

Microscopy imaging is widely used to capture volumetric biological structures through z-stack acquisition. A common approach is to project these stacks into a single 2D image using maximum intensity projection (MIP) [1]. However, during acquisition, physical factors such as stage drift and focus drift degrade image quality.

XY drift introduces spatial misalignment across slices, while Z drift leads to defocus blur due to out-of-focus imaging [2]. These effects reduce image fidelity and negatively impact downstream analysis. Traditional projection methods such as MIP do not explicitly correct for these degradations.

In this project, we simulate physically motivated imaging degradations and train a neural network to reconstruct clean fused images from distorted z-stacks. The task is defined as learning a mapping from degraded stacks to the ideal MIP result.

2 Related Work

Deep learning has been widely applied to image restoration tasks in biomedical imaging, including denoising, deblurring, and super-resolution. Convolutional neural networks (CNNs), particularly U-Net [3], have been extensively used due to their ability to capture both local and global image features.

In microscopy imaging, several studies have applied deep learning to improve image quality. For example, Weigert et al. [4] proposed Content-Aware Image Restoration (CARE), which trains neural networks using paired degraded and clean images for denoising and restoration tasks. Similarly, other approaches use supervised learning to map corrupted microscopy images to high-quality targets.

These methods typically rely on paired training data, where degraded inputs and corresponding clean images are provided. In many cases, the degradation process is either experimentally obtained or synthetically generated, and the neural network learns a direct mapping between the two domains.

Separately, there has been increasing interest in incorporating physical knowledge into machine learning models. Physics-informed neural networks (PINNs) [5] and related approaches integrate governing equations or forward models into the training process to improve interpretability and generalization.

In this project, we adopt a simulation-based approach, where physically motivated degradations are explicitly modeled to generate training data. However, the neural network itself is trained using a standard supervised learning framework without explicitly embedding the physical model into the learning objective.

3 Methods

3.1 Dataset and Baseline

We used the `cells3d` dataset from `scikit-image` [6], which provides volumetric fluorescence microscopy images with multiple z-slices.

As a baseline fusion method, we applied maximum intensity projection (MIP), which generates a 2D image by selecting the maximum intensity value along the z-axis at each pixel location. The MIP result serves as the clean ground truth for reconstruction, as shown in Figure 1.

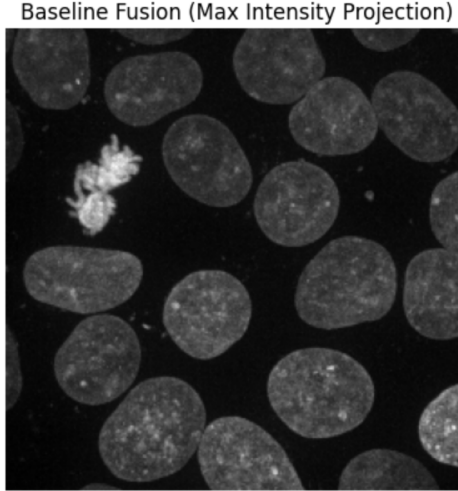


Figure 1: Example of clean MIP generated from a z-stack.

3.2 Physical Degradation Modeling

To simulate realistic imaging conditions, we introduce two types of physically motivated degradation that capture common sources of error in microscopy acquisition.

First, XY drift is modeled as a sequence of small random translations applied to each slice, where the displacement accumulates across the z-stack. The total displacement is bounded by a maximum of 6 pixels. This formulation reflects the fact that stage drift in real imaging systems is not independent across slices, but instead evolves gradually over time, leading to cumulative spatial misalignment.

Second, Z drift is modeled as depth-dependent defocus blur. Specifically, the middle slice is treated as the focal plane, and the amount of Gaussian blur increases with distance from this plane. The maximum blur level is controlled by a standard deviation of $\sigma = 2.0$, which approximates the loss of high-frequency information in out-of-focus regions.

For the combined degradation setting, Z drift is applied first to simulate defocus, followed by XY drift to introduce spatial misalignment. This ordering reflects the sequential nature of these effects during image acquisition and the result is shown as Figure 2.

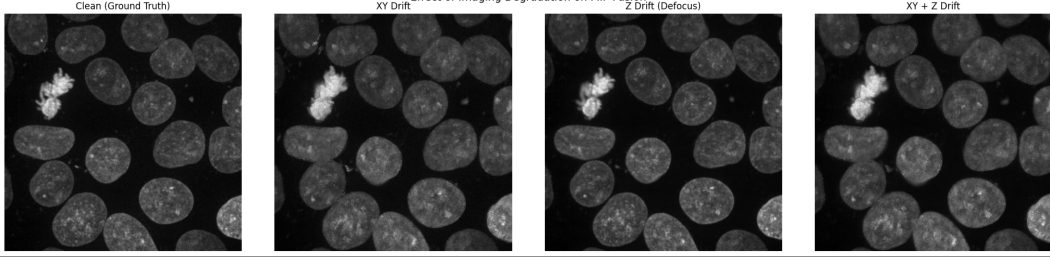


Figure 2: Effect of different types of drift on MIP fusion.

3.3 Patch-Based Data Generation

Training data are constructed by extracting local patches from the simulated degraded z-stacks and the corresponding clean MIP image. For each degraded stack, a sliding window is applied across the spatial dimensions. Each input sample is a z-stack patch with shape $(Z, 32, 32)$, where Z is the number of slices in the stack. The corresponding target is the clean MIP patch from the same spatial location with shape $(1, 32, 32)$.

Patches are extracted using a stride of 8 pixels. To generate multiple degraded examples, five distorted stacks are created from the same clean stack using different random seeds. After all patches are collected, the dataset is split into training and validation sets using an 80/20 random split at the patch level.

Because the validation patches are sampled from the same pool of simulated degradation settings, this validation strategy evaluates performance on held-out patches rather than fully independent drift distributions. This limitation is considered in the discussion.

3.4 Model Architecture

We use a lightweight U-Net model to reconstruct the clean fused image from each degraded z-stack patch. The input to the network is treated as a multi-channel image, where each z-slice corresponds to one input channel. The network predicts a single-channel reconstructed MIP patch.

The model follows an encoder-decoder structure with skip connections. Each convolutional block consists of two 3×3 convolutional layers, each followed by a ReLU activation. The encoder contains two feature levels with 32 and 64 channels, separated by 2×2 max pooling. The bottleneck uses 128 channels. In the decoder, transposed convolutions are used for upsampling, and the upsampled features are concatenated with the corresponding encoder features through skip connections. A final 1×1 convolution followed by a sigmoid activation produces the reconstructed image.

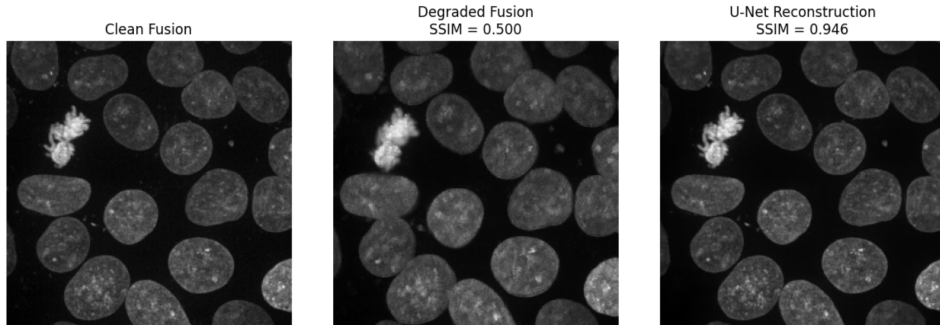


Figure 3: Qualitative reconstruction results using the U-Net model.

3.5 Training Details

The model is trained using the Adam optimizer with a learning rate of 1×10^{-3} , a batch size of 16, and 180 training epochs. During each epoch, the model is trained on mini-batches from the training set and evaluated on the validation set.

The training objective combines a pixel-wise reconstruction loss with an edge-aware loss:

$$\mathcal{L} = \mathcal{L}_{L1} + 0.2 \cdot \mathcal{L}_{edge}. \quad (1)$$

Here, $\mathcal{L}_{L1}$ is the mean absolute error between the predicted patch and the clean MIP target. The edge term is computed by applying a discrete Laplacian filter to both the prediction and the target, then taking the mean absolute difference between the two Laplacian responses. This term encourages the reconstructed image to preserve high-frequency structures, while the smaller weight prevents it from dominating the pixel-wise reconstruction objective.

4 Results

4.1 Quantitative Evaluation of Physical Degradation

We first evaluate the impact of simulated drift on image quality using two complementary metrics: structural similarity index (SSIM) [7] and Laplacian variance. SSIM measures the preservation of structural information relative to the clean MIP image, while Laplacian variance is used as an estimate of image sharpness, reflecting the amount of high-frequency content.

Figure 4 shows the SSIM values under different degradation conditions. XY drift leads to a significant reduction in SSIM due to spatial misalignment between structures, while Z drift has a smaller effect since the overall spatial arrangement remains largely preserved despite blurring. The combined degradation produces the lowest SSIM, reflecting the combined impact of misalignment and blur.

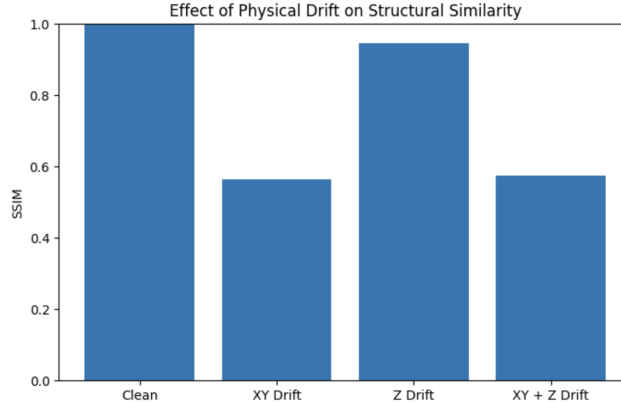


Figure 4: SSIM comparison under different degradation conditions. XY drift significantly reduces structural similarity due to misalignment, while Z drift has a smaller effect.

Figure 5 shows the corresponding Laplacian variance values. In contrast to SSIM, XY drift has a relatively small effect on Laplacian variance, as it mainly shifts structures without removing edges. Z drift, however, leads to a clear decrease in Laplacian variance, indicating loss of high-frequency detail due to defocus blur. These results confirm that SSIM and Laplacian variance capture different aspects of image degradation.

4.2 Quantitative Evaluation of Reconstruction Performance

We next evaluate the reconstruction performance of the U-Net model using the same two metrics.

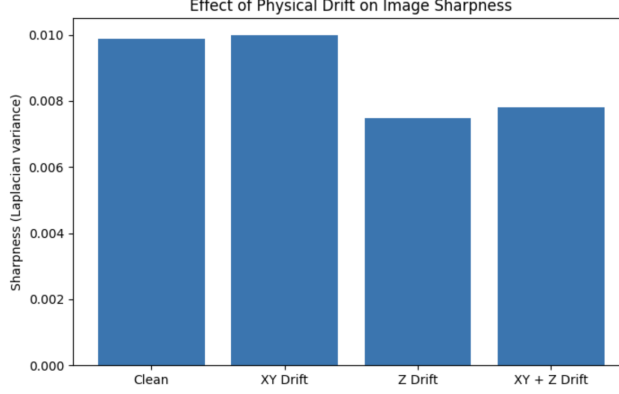


Figure 5: Laplacian variance under different degradation conditions. Z drift significantly reduces sharpness, while XY drift has a smaller effect.

Figure 6 shows the SSIM values before and after reconstruction. The degraded input has an SSIM of approximately 0.51, which increases to approximately 0.95 after reconstruction. This indicates that the model successfully restores global structural information.

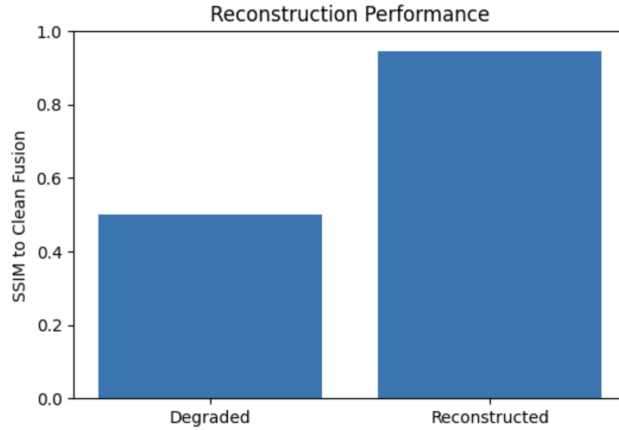


Figure 6: SSIM comparison between degraded input and reconstructed output. Reconstruction significantly improves structural similarity.

Figure 7 shows the corresponding Laplacian variance values. While reconstruction improves sharpness compared to the degraded input (from ~ 0.0084 to ~ 0.0092), it does not fully recover to the level of the clean image (~ 0.0110). This suggests that the model partially restores high-frequency content but still produces smoother outputs than the ground truth.

5 Discussion

The results demonstrate that the proposed approach is effective in recovering structural information from drift-degraded z-stacks. The significant improvement in SSIM (from ~ 0.51 to ~ 0.95) indicates that the model successfully reconstructs overall spatial structure and intensity relationships. This suggests that a supervised learning framework can learn a meaningful mapping from degraded stacks to clean MIP images, even when the degradation process is relatively complex.

However, the reconstruction quality remains limited in terms of high-frequency detail recovery. The Laplacian variance does not fully return to the level of the clean image, and the reconstructed outputs appear slightly over-smoothed. This behavior is consistent with the choice of loss function. The L1 loss dominates the optimization objective and encourages pixel-wise averaging, which often leads to

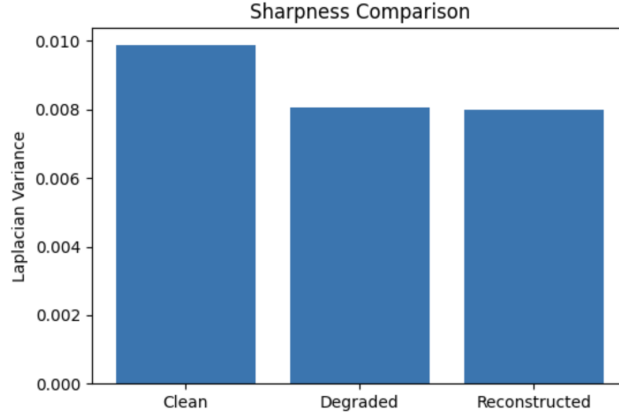


Figure 7: Laplacian variance comparison for reconstruction. Sharpness is partially recovered but remains below the clean reference.

smoother results. Although an edge-aware Laplacian loss is included, its relatively smaller weight (0.2) limits its influence on preserving fine structures.

Another important observation comes from analyzing the effects of different types of drift. XY drift primarily affects spatial alignment, leading to reduced structural similarity (SSIM), while having a relatively small effect on image sharpness. In contrast, Z drift introduces defocus blur, significantly reducing high-frequency content and thus lowering Laplacian variance. These results confirm that the simulated degradation model captures distinct physical effects that influence image quality in different ways.

Despite these promising results, the model exhibits limited generalization capability. Since the degraded data are generated using fixed simulation rules (e.g., bounded random shifts and predefined blur profiles), the model tends to learn specific distortion patterns rather than a fully generalizable inverse mapping. Furthermore, because validation is performed using held-out patches sampled from the same simulated degradation settings, this experiment does not fully test generalization to unseen drift distributions. In real microscopy systems, drift is stochastic and varies across acquisitions, making it difficult for a model trained on a limited set of simulated distortions to generalize.

This highlights an important distinction between simulation-based training and more advanced approaches that explicitly incorporate physical models into the learning process. In this project, the physical model is used only to generate training data, while the neural network itself remains a standard supervised model. As a result, the learned mapping is empirical rather than physically constrained.

Future work could improve performance in several ways. First, increasing the diversity of simulated distortions, such as varying drift magnitude, direction, and blur profiles, may improve generalization. Second, incorporating multiple datasets could help the model learn more robust features. Finally, alternative loss functions, such as perceptual or frequency-based losses, could improve the recovery of high-frequency details.

References

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